Historic-Log Controls and Active-Assay Preconditions

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Abstract

This working supplement specifies conventional identification controls for one historical cascaded-tanks record pair. A complete encoder state-space endpoint retains all three optimization repeats; seed sensitivity and validation improvement through the training cap limit its interpretation. A frozen linear/quadratic pilot retains its adverse longer-simulation outcome, and its required two-family comparison remains withheld. A source-derived fixed-light ambiguity and a conditional growth-surrogate action distinction explain why a future calibration assay needs informative, explicitly costed interventions and a task-specific teacher response. The supplement establishes neither allocator efficacy nor physical deployment.

1. Historic-log identification controls

Observable scope. The cascaded-tanks benchmark provides an estimation record and a separate test record, each with 1,024 input/output samples at a four-second sampling period (Schoukens & Noël, 2017). The measured quantities are pump input and lower-tank sensor voltages. Upper-tank state, calibrated level, flow, dose and complete overflow labels are unavailable. We obtained the archive from an author-listed fallback; byte equivalence to the unavailable owner archive is unproved. These records support retrospective prediction, not active interventions, resets, teacher queries or physical qualification.

Temporal selection and isolated test scoring. Three estimation-only folds fit prefixes of length 512, 640 and 768 and evaluate the immediately following 128 samples. Each recursive evaluation starts from 50 observed input/output pairs and then uses supplied inputs without evaluation-output feedback. Final models use the entire estimation record. Test prediction receives all 1,024 known inputs and only the first 50 measured outputs. All required models are sealed before prediction and all predictions before a separate scorer accesses the 974-sample suffix. The model APIs separate these roles; this is not an operating-system access-control claim. Figure 1 shows the calendar.

Lag-only linear and quadratic models. For history length p, let $z_t = (\widetilde{u}_{t-1:t-p}, \widetilde{y}_{t-1:t-p})$, where normalization uses only the current training prefix. Linear features are $(1, z_t)$; quadratic features add all products $z_{t,i} z_{t,j}$ with $i \le j$. Recursive prediction substitutes preceding model outputs for unobserved output lags. Contemporaneous input u_t is omitted from both classes. The frozen grid comprises degree 1 or 2, history $p \in \{2, 4, 8, 16\}$ and ridge penalty $\lambda \in \{10^{-6}, 10^{-4}, 10^{-2}, 1, 100\}$: 40 candidates and 120 fold fit attempts. Within each family, selection minimizes the mean of all three finite voltage RMSE values, with exact ties resolved by smaller history and then larger penalty. A failed constituent makes the candidate unavailable; no surviving-fold denominator is substituted. Selected models receive one full-estimation refit each. Outputs are never clipped or repaired.

Encoder state-space control. A separate conventional adaptation of Beintema et al. (2021) learns

$$\begin{aligned} x_{t_0} &= E_\theta(u_{t_0-50:t_0-1}, y_{t_0-50:t_0-1}), \\ x_{t+1} &= F_\theta(x_t, u_t), \\ \widehat{y}_t &= H_\theta(x_t, u_t). \end{aligned}$$

Here t_0 is the initialization index. The encoder is followed by free simulation; it is not repeatedly reinitialized from unknown outputs. Each network has a 32-unit tanh hidden layer and a linear bypass. Latent dimension is 2 or 4; coordinates are learned representations, not calibrated tank levels. Training uses CPU float32 arithmetic, deterministic operations and one numerical thread; Adam has learning rate 10^{-3}, no weight decay and gradient-norm cap 5. Each update samples 64 sections with a 50-pair encoder history and 64-step simulation target. Gradient clipping does not clip predictions.

Each dimension, seed and fold receives 10,000 updates: 18 training runs, with checkpoints at 1,000, 3,000 and 10,000 updates. Six dimension/checkpoint candidates are compared using all nine fold/seed constituents, for 54 checkpoint evaluations. Nonfinite or missing constituents make a candidate

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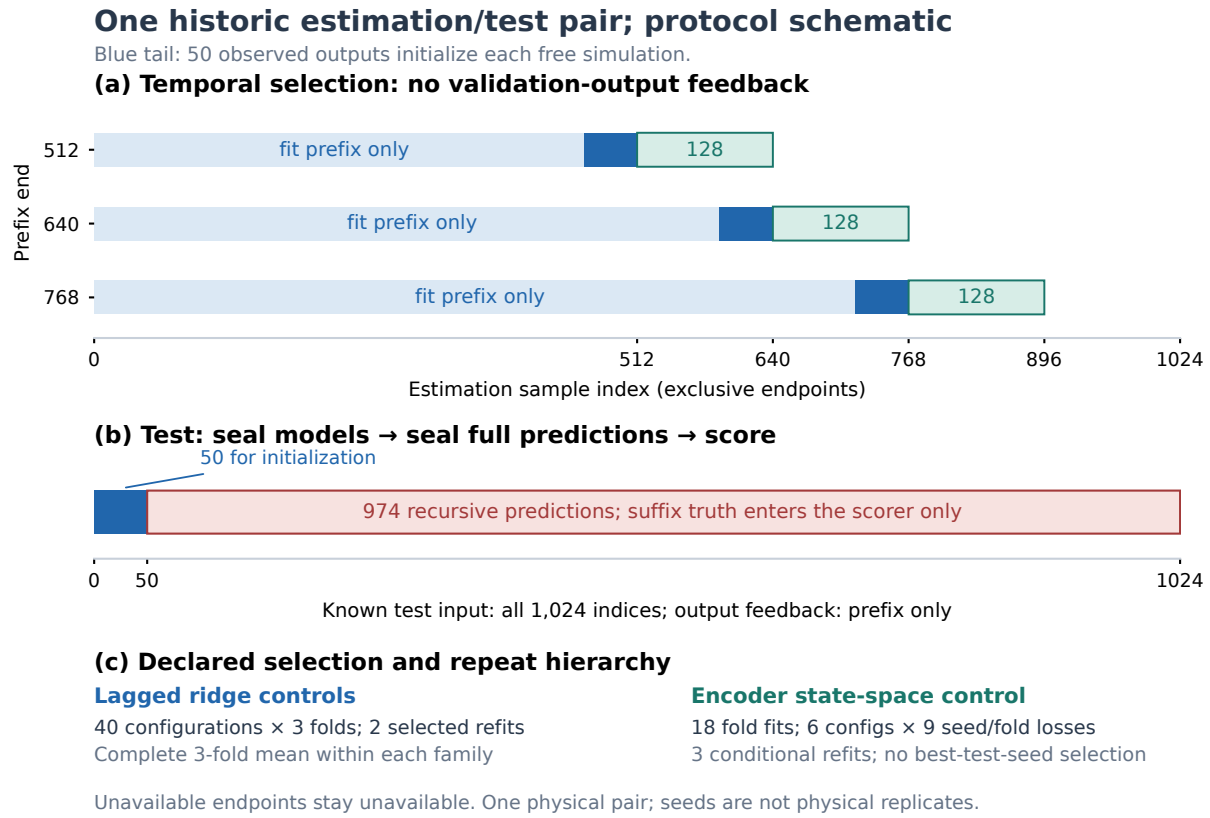


Figure 1. Frozen historic-log protocol. Blue denotes permitted observation or initialization; green denotes estimation-only validation; red denotes the separately scored test suffix. Neither validation nor test suffix outputs feed a recursive prediction. The two controls have different input information and optimization budgets. Required failures remain visible, and complete endpoints require genuine terminal status and independent result review. The schematic contains no measured results.

unavailable; exact ties favor smaller dimension and then earlier checkpoint. The chosen structure/checkpoint is refit on the full estimation record for all three seeds, with no best-test seed selection. The total cap is 210,000 updates, or 860,160,000 simulated training output steps, not an equal-FLOP or equal-ridge-fit budget. The three seeds are optimization repeats of one physical record pair.

A separately frozen diagnostic initializes saved selected-prefix checkpoints from samples $[462, 512)$ and simulates $[512, 1024)$, a 512-step estimation-record horizon. It adds no fit, selection rule or test exposure; it is not a new holdout. All three final 1,024-sample predictions must be finite before the three 974-suffix scores constitute the defined complete comparator endpoint. The sole attempt and its independent result review are complete, within the one-record-pair scope reported below. Reaching the training cap does not establish convergence. Neither this adaptation nor comparison with the earlier lag-only model is an equal-information reproduction of published results.

Table 1. Complete conventional endpoint, in sensor volts. Every declared optimization seed is retained. Test and estimation-only diagnostic horizons differ; their numbers are not a matched comparison. Mean and range describe optimization variation on one record pair, not physical replications or confidence intervals.

Seed	Test (974)	Diagnostic (512)
96200	0.3773	0.5092
96201	0.7252	1.2987
96202	0.7311	0.6861
Test mean	0.6112	—
Test range	[0.3773, 0.7311]	—

2. Complete conventional state-space endpoint

All 26 declared phases completed successfully. All 54 checkpoint evaluations were finite, and every candidate retained its nine seed/fold losses. Selection chose dimension 4 and 10,000 updates, with validation RMSE 0.5121 volts. Each final model has 4,347 parameters; all three seeds are retained.

Table 1 retains all three 974-suffix scores; their descrip-

tive mean is 0.61119854 volts. Independent review recomputed saved-array residuals and scores and verified complete sources, constituents and chronology, without refitting, predicting or decoding the original MAT. The measured initialization prefix is excluded from scoring.

Actual training across 21 runs used 210,000 updates, 13,440,000 sampled sections and 860,160,000 forward target/output steps. Optimizer CPU time summed to 2,564.67 seconds; complete owner wall time was 3,326.23 seconds, including other phases and supervision. These different scopes are neither physical interactions nor FLOPs or an equal-ridge-fit budget. Selection added 54 128-point simulations; the nonselecting diagnostic and final prediction added three 512-point and three 974-point simulations, respectively, without additional fits.

Optimization and competence limits. Validation improves through the update cap for both dimensions (Figure 2a). A post hoc comparison of ten batch-loss observations at updates 8,100–9,000 with ten at 9,100–10,000 found a lower late mean in 19 of 21 runs. These stochastic batches are neither matched full-objective evaluations nor a preregistered convergence test. Finite prediction under the declared budget establishes neither convergence nor a fully optimized conventional control. Seed sensitivity and long-rollout errors remain visible. Later architecture or budget changes require a separate declaration on this exposed benchmark. This endpoint cannot repair the failed two-family comparison below; different current-input access and training budgets preclude a superiority inference.

3. Complete adverse outcome of the lag-only pilot

The sole frozen pilot completed all four phase commands and preserved all 120 fold-fit attempts and two selected full-estimation fits. Of the 120 validation records, 114 have finite complete 128-sample scores and six quadratic records failed: five recursive nonfinite predictions and one nonfinite score. All 20 linear candidates and 15 of 20 quadratic candidates have complete finite three-fold means. Failed candidates were retained and excluded by the declared completeness rule, without replacement losses.

Both selected full-estimation models have finite coefficients. The quadratic model's three selected validation segments are finite, yet its subsequent requested 974-step test suffix simulation becomes nonfinite. Its prediction and score are null. The first nonfinite index and completed recursive prefix were not recorded and remain unknown; 974 denotes the requested horizon, not completed quadratic steps. The original command therefore terminated with failure and the complete comparison was withheld. Independent review verified the preserved phase/input/output records and recomputed the linear suffix RMSE, 0.6161362115512835 sensor volts, directly from sealed prediction and truth arrays without refitting or resimulation.

Table 2. Selected pilot models from one historic record pair. Validation RMSE is the mean of three complete 128-step folds, in sensor volts. The linear test number is a retained partial diagnostic. The required two-family test comparison is withheld.

	Linear	Quadratic
History p	16	8
Ridge λ	10^{-6}	10^{-6}
Coefficients	33	153
Validation RMSE	0.4071	0.4162
Scored test samples	974	0
Test RMSE	0.6161	unavailable

This record shows that finite short validation and finite fitted coefficients did not ensure finite longer free simulation for the selected quadratic model. It does not estimate an instability rate or establish a physical cause. The linear diagnostic does not become an accepted two-family result by omitting its failed counterpart. No clipping, test-conditioned grid change, additional refit or model substitution was performed. The state-space protocol was frozen before this adverse outcome. One record pair supports neither population confidence intervals nor independent physical replication.

4. An informative-intervention requirement

Candidate model and conventional prior art. The published photoproduction task (Pan et al., 2021) and the public PC-Gym framework (Bloor et al., 2024) offer a possible future interactive setting. Online model updating and uncertainty-aware control already appear in Bradford et al. (2020). A prospective calibration-to-query assay must therefore include a competent conventional updated-model control; online updating alone is not the proposed contribution. The following argument uses the published growth model and does not execute the candidate simulator.

Fixed-light ambiguity. For positive light I and known nitrate concentration $N > 0$, the model's growth factor is

$$g(I, N) = \frac{I}{I + k_s + I^2/k_i} \frac{N}{N + k_N}.$$

At one fixed light $I_0 > 0$, parameters k_s, k_i enter only through $s_0 = k_s + I_0^2/k_i$. For any nearby positive k_i' satisfying positivity of

$$k_s' = k_s + I_0^2(1/k_i - 1/k_i'),$$

the same s_0 and growth factor result. With other parameters, initial conditions and nitrate feed fixed, biomass and nitrate derivatives are identical. The product equation has no direct

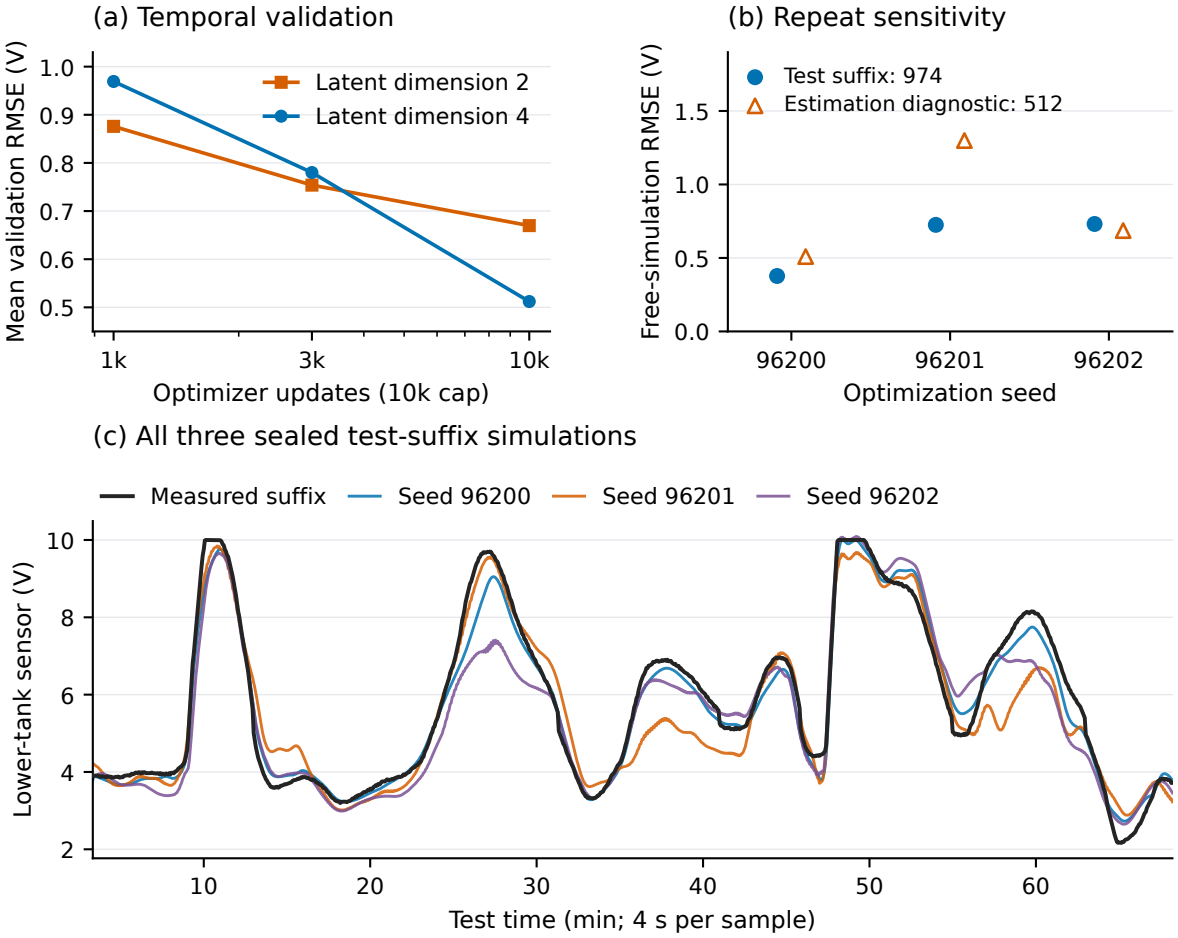


Figure 2. Complete conventional result on one historical record pair. (a) Validation means retain all nine fold/seed constituents and continue decreasing through the training cap. (b) All three optimization repeats are shown for the 974-sample test suffix and the nonselecting 512-sample estimation-only diagnostic; horizons and information scopes differ. (c) All three original sealed test-suffix predictions and measured truth, without clipping, smoothing or best-seed selection. Test truth entered the scorer after predictions were sealed. This figure reports no allocator effect, new-method advantage or convergence guarantee.

k_s, k_i dependence, so its derivative is identical as well. On their common domain of unique ODE solutions, the resulting noiseless trajectories coincide. A parameter-independent observation-noise law cannot remove this ambiguity.

For a different light, the growth denominator changes by

$$(I^2 - I_0^2)(1/k_i' - 1/k_i).$$

Thus parameters indistinguishable from a fixed-light record can differ under a later intervention. This is a possible source of action-relevant ambiguity, not a proof that a teacher changes its optimal action or that calibration improves control.

Ideal local information and its limits. With common known N and exact instantaneous growth observations,

$$I/g(I,N) = rk_s + rI + (r/k_i)I^2, \qquad r = 1 + k_N/N.$$

Three distinct positive light levels identify quadratic coefficients a_0, a_1, a_2. Under the positive-parameter model, $k_s = a_0/a_1$, $k_i = a_1/a_2$ and $k_N = N(a_1 - 1)$. This three-parameter statement assumes exact normalized growth, known growth scale and known decay. Extracting g from measured biomass also requires positive biomass and exact derivatives. At common N, an unknown growth scale is confounded with the nitrate factor, so this argument does not identify k_N without that scale. It does not supply exact derivatives from noisy 20-hour observations, cost-free

matched states or a physical reset, and does not prescribe a practically sufficient sample count. Feed variation at one fixed light cannot resolve the k_s/k_i continuum. An implemented calibration service must expose and cost informative light variation, or infer only identifiable combinations.

A conditional surrogate action distinction. Let $D = I + k_s + I^2/k_i$ and $q = N/(N + k_N) > 0$. Then

$$\frac{\partial g}{\partial I} = q\frac{k_s - I^2/k_i}{D^2}.$$

Instantaneous growth has the unique unconstrained maximizer $I_g^* = \sqrt{k_s k_i}$; light bounds project it onto the allowed interval. For the fixed-light-equivalent pair above, if both optima lie in the interval's interior,

$$(I_g'^*)^2 - (I_g^*)^2 = (k_i'/k_i - 1)(k_s k_i + I_0^2).$$

Thus a nontrivial equivalent pair has different growth-surrogate actions. For nominal $k_s = 178.9$, $k_i = 447.1$, $I_0 = 200$ and $k_i' = 1.1k_i$, the growth optima are approximately 282.8183 and 303.2900, both interior to $[120, 400]$. An exact-arithmetic witness verifies the algebra. This deterministic example is not a draw from the unresolved uncertainty law. Projection can make both actions identical at a shared boundary.

The distinction is objective-dependent. In the same source model, the instantaneous product-rate light factor uses the separate constants k_{sq} and k_{iq}. Holding state and those constants fixed gives the same light maximizer $\sqrt{k_{sq}k_{iq}}$ for both growth-parameter pairs. Neither this counterexample nor the growth-surrogate result solves the finite-horizon terminal-product teacher. A positive unknown common growth scale does not change the instantaneous growth argmax; it still matters for the earlier three-parameter identification claim. These results motivate an actual task-specific teacher-response assay, rather than establishing calibration benefit or an allocator effect.

Open conditions for a complete mechanism assay. Static inspection of a pinned framework revision leaves parameter propagation to the actual transition engine unverified. Its sampled parameters can remain appended to the returned observation, while metadata copies an unmasked observation before masking. A future shared collector must project permitted physical coordinates into a fresh public record and must not forward raw metadata. A pure projection spike tests selected-only reads, nonpublic-coordinate and metadata invariance, mask preservation and detached serialization. Its guarantees require a trusted layout and sole reviewed return; arbitrary direct record construction, a false layout, raw object access and other channels remain outside that contract. It does not implement the framework adapter or authenticate physical units, context or causal sequence. The reset seed argument alone does not establish campaign randomization. The published parameter variance description and implementation's standard deviation comment disagree; the uncertainty law is unresolved. These are source-derived concerns, not runtime reproductions or validated priors. Before an effect assay, the project must establish transition binding, measured return interfaces, campaign/reset costs, competent teacher responses conditioned on the actual updated model and learner state, development headroom and equal information rights for all controls. The main mechanism and full scientific acceptance remain open.

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